

# **Informed Consent and Institutional Liability in Participation Systems Without Standardized Outcome Disclosure or Safety Modeling**

## **Executive summary**

Participation systems (organized activities in which individuals “opt in” to roles, tasks, exposure, or data processing under rules set by an institution or platform) routinely rely on consent—sometimes as a legal authorization (e.g., bodily interventions, data processing), sometimes as a defense narrative (assumption of risk), and sometimes as a governance claim (“participants agreed”).

Across US, EU, and UK legal regimes, the absence of standardized outcome disclosure (reliable, comparable reporting of relevant incident rates, adverse events, or other material outcomes) and the absence of safety modeling (formal risk assessment/risk management processes that anticipate and mitigate foreseeable harms) do not automatically “invalidate consent” in every domain. But they frequently weaken or defeat (i) the conditions for legally “informed” consent where consent is a precondition of lawfulness, and/or (ii) defenses that shift responsibility onto individual participants (waivers, assumption-of-risk), and they can materially raise institutional and platform exposure under negligence, product liability, consumer protection, data protection, and sector-specific online-safety frameworks.

The most consistent comparative finding is this: when harms are documented, structurally recurrent, and foreseeable, modern regimes increasingly treat the absence of risk assessment, monitoring, and transparent reporting as evidence of breach (failure to warn, negligent design, inadequate governance controls) and as a basis to reject “participant choice” narratives—especially where the organizer/platform controls the environment, holds decisive information, or uses interface design to shape behavior.

In regulated “high-duty” settings (human-subjects research and certain online-safety regimes), the shift is more explicit: regulators require risk assessments, record-keeping, and in some

cases public-facing summaries. Failure can trigger administrative penalties and remedial orders, and can also change civil litigation dynamics by undermining “voluntary assumption” arguments and by supplying per se / statutory-breach theories where available.

## **Scope, assumptions, and definitions**

This report responds to: “In participation systems where harms are historically documented, structurally consistent, and foreseeable, does the absence of standardized outcome disclosure and safety modeling invalidate informed consent and shift liability from individual participants to organizing institutions and platforms?”

Because “participation system” is underspecified, the analysis assumes a broad class of systems with shared structure: (i) the organizer/platform defines rules and interface, (ii) participation is opt-in but often shaped by information asymmetry and design nudges, and (iii) participation produces exposure to physical/psychological/economic/data harms.

### **Assumptions (explicit)**

The conclusions below are conditioned on these default assumptions unless otherwise specified:

- Participants include adults and may include minors (which elevates duties in many regimes).
- The organizer/platform has superior information about incident rates, risk pathways, or design impacts (e.g., internal telemetry, moderation logs, safety testing).
- Harms are “historically documented” via credible sources (peer-reviewed evidence, regulator reports, litigation records, or robust surveillance data), and are “structurally consistent” in the sense that they recurrently emerge from stable system features rather than isolated freak events.
- The relevant decision is whether consent is legally effective (for legality of conduct/data processing), and/or whether consent/waiver meaningfully supports organizer defenses (assumption of risk, contributory negligence, contractual releases).

## Working definitions

Participation system: A socio-technical and/or institutional arrangement in which individuals participate under an organizer/platform's designed environment (rules, interfaces, incentives), generating exposure and outcomes, with participation justified or structured through notice/consent, terms, or enrollment documents. (Examples: clinical research participation, platform-mediated gig work, large social media platforms, contact sports programs, subscription/digital consumer interfaces.)

Standardized outcome disclosure: Systematic reporting of relevant outcomes (e.g., adverse events, injury rates, complaint rates, content/harm prevalence, risk-level summaries) in a format that is comparable across time or across services and is understandable enough to inform participation decisions or oversight. In some regimes this includes public reporting obligations or mandated summaries (e.g., certain online-safety transparency reporting and risk assessment summaries).

Safety modeling: Formalized risk assessment and mitigation processes (hazard identification, severity/likelihood estimation, controls, auditing, periodic review, "safety by design"), sometimes mandated (online safety risk assessments; high-risk AI risk management; research ethics risk/benefit assessment).

## Legal analysis across jurisdictions

### Comparative table of legal standards by jurisdiction

The table synthesizes how "informed consent," disclosure, and risk modeling interact with liability. It focuses on the most analytically relevant legal contexts for the posed question.

| Topic | US | EU | UK | |--|--|--| | Consent as a condition of legality (core domains) | In human-subjects research, federal regulations require consent that provides "information that a reasonable person would want," with disclosure of "reasonably foreseeable risks," and prohibit "exculpatory language" waiving rights or releasing investigators/institutions from negligence. In clinical contexts, courts frame informed consent as grounded in bodily autonomy and a duty to disclose material risks/alternatives. | Under GDPR, lawful consent must be "freely given,

specific, informed and unambiguous.” EU supervisory and advisory bodies treat manipulative design (“dark patterns”) as capable of undermining valid GDPR consent. | In medical law, the Supreme Court’s disclosure standard is patient-centered (“material risks” and reasonable alternatives), emphasizing autonomy. Some disclaimer/waiver strategies are limited by statute for negligence causing death/personal injury. || Role of “standardized outcome disclosure” in consent validity | Not universally required outside regulated areas; but where law requires disclosure of foreseeable risks, failure to disclose outcome distributions or known adverse event history (when material) can vitiate informed decision-making and support liability theories. Clinical trial results reporting is legally mandated in defined circumstances (results disclosure/registration regime), supporting sectoral standardization and oversight. | GDPR’s “informed” requirement implies meaningful transparency about processing purposes and consequences; EU institutions treat deceptive interface patterns as threatening valid consent. The DSA adds transparency and systemic-risk reporting duties for very large platforms/search engines (including making risk assessment results available and annual reporting). | Online Safety regime creates structured disclosure ecosystem: risk assessment guidance, required record-keeping, and for certain categories of services, publication of risk assessment summaries tied to terms or public statements (including levels of risk and severity). || Safety modeling / risk assessment duties | General US tort law can treat failure to test, monitor, warn, or design safer systems as negligent depending on context; in regulated research, risk/benefit framing is foundational in ethics and compliance. In platform consumer contexts, agencies emphasize that deceptive design can negate meaningful choice, pointing toward a functional governance duty to evaluate and mitigate predictable harms. | DSA requires annual systemic risk assessments for very large platforms/search engines, including assessment of “any actual or foreseeable negative effects” and consideration of recommender systems, terms and conditions, and other factors; it also requires risk mitigation measures and reporting. The AI Act imposes risk-based obligations and explicitly recognizes AI’s capacity to cause physical, psychological, societal, or economic harm, motivating risk management obligations for high-risk systems. | Online Safety Act duties are operationalized via Ofcom’s guidance: “suitable and sufficient” risk assessment standard; required consideration of service characteristics, algorithmic systems, intended/unintended use; and enforcement penalties for failure. ||

Liability shift logic (high level) | Absence of disclosure/modeling typically does not “shift liability” automatically, but it can defeat assumption-of-risk narratives and support negligence/product-design claims, especially where harm is foreseeable and institution controls design/conditions. | Increasingly, EU law builds duties into platforms and AI/product ecosystems via ex ante risk assessment and transparency obligations. Breach increases enforcement exposure and can influence civil liability through product liability and national tort principles (varies by Member State). | UK: failure to disclose material risks in clinical context is actionable negligence; statutory constraints limit exclusion of liability for death/personal injury due to negligence; online-safety regulation imposes structured risk assessment and record-keeping providing a compliance benchmark relevant to responsibility allocation. |

## **Legal “hinge points” that determine whether consent is invalidated and liability moves**

### **Whether “consent” is operating as legal authorization or as a defense narrative**

A central analytical move is to distinguish two “workings” of consent:

- Consent as legal authorization: In domains like human-subjects research, consent is a compliance condition and must be “legally effective,” include reasonable-person information and foreseeable risks, and must not contain exculpatory language releasing institutions from negligence. In EU data protection law, consent must be “informed” and “unambiguous,” and deceptive designs can compromise that validity.
- Consent as liability-shield narrative: In consumer/platform and recreational contexts, “consent” is often operationalized as (i) a waiver/release clause, (ii) terms acceptance, and/or (iii) assumption-of-risk framing. These are more defeasible when users lack material risk information, when public policy limits releases, or when law prohibits exclusion for certain harms.

This distinction matters because the posed question asks whether disclosure/modeling absence “invalidates informed consent.” In authorization settings, that can be close to literally true; in defense-narrative settings, the better framing is that disclosure/modeling absence often weakens or defeats the organizer’s defenses and strengthens duty/breach arguments.

## **The “materiality” of undisclosed outcomes and the organizer’s epistemic advantage**

US informed consent doctrine emphasizes that the patient’s ability to choose “knowledgeably” depends on disclosure of risks whose significance is shaped by both incidence and severity, and links liability to causation via a “prudent person” standard. This logic creates a pathway: where outcome distributions are known (or should be known) and would influence a reasonable participant’s decision, failing to disclose them undermines informed decision-making and supports liability.

Regimes that require safety modeling operationalize the same intuition: Ofcom expects risk assessments to consider service characteristics, user base, algorithmic systems, business model, intended/unintended uses, and to estimate likelihood and impact; it explicitly expects evidence inputs (core and “enhanced” evidence such as product testing outputs, audits, and consultation) and treats failure as enforceable. The DSA similarly requires annual systemic risk assessments that include “foreseeable negative effects” and explicitly calls out recommender systems, content moderation, and terms/conditions as risk drivers.

Taken together, these sources support a cross-jurisdictional claim: where organizers have superior information and control system design, the law’s direction of travel is to treat a lack of robust risk assessment and outcome reporting as a due care failure, not as a participant’s “informed choice.”

## **Causation and “proximate” limitations**

Even when disclosure failures are established, liability allocation still hinges on causation doctrines that differ by domain:

- In US informed consent negligence framing, causation is often analyzed via whether a prudent person would have declined the intervention if adequately informed.
- In UK medical negligence, courts have treated a physician’s failure to warn as specially connected to autonomy interests and have addressed causation in a distinct way in some cases, reflecting sensitivity to “what consent protects.”
- In platform harms, proximate cause arguments (intervening choice, third-party acts) often dominate; but where claims target design features rather than publishing

third-party speech, US courts have allowed suits to proceed notwithstanding online intermediary immunity arguments in at least some circumstances.

## **Ethical analysis**

Ethically, the posed question tracks the Belmont framework: respect for persons implies meaningful autonomy (not formalistic signatures), beneficence implies systematic risk assessment and risk minimization, and justice implies attention to structural vulnerability and differential burdens.

A key ethical implication is that unstandardized disclosure can convert “consent” into a procedural artifact rather than a genuine authorization. Modern consent standards also emphasize readability and comprehension: US human-subjects regulations require a “concise and focused” presentation of key information and prohibit exculpatory language that waives rights or releases institutions from negligence.

Ethics also directly connects to interface and governance design. The Federal Trade Commission describes “dark patterns” as design practices that can trick or manipulate consumers into decisions they would not otherwise make, highlighting buried terms, difficult cancellation, and deceptive framing as recurring tactics. The European Data Protection Board similarly treats deceptive design patterns in social media interfaces as incompatible with GDPR-aligned user control and meaningful consent.

The ethical upshot is aligned with the legal trend: when systems predictably generate harm, the ethical burden shifts from “did the participant click agree?” to “did the organizer model, mitigate, and disclose risks in a way that a reasonable participant can understand and that respects autonomy?”

## **Empirical evidence of documented harms**

The question conditions analysis on harms that are historically documented, structurally consistent, and foreseeable. The empirical record supports that many participation systems have recurrent harm mechanisms tied to stable features such as incentive structures,

algorithmic amplification, interface design, exposure intensity, and risk compression onto individuals.

## Comparative table of documented harms and evidence strength

Evidence strength is context-dependent; here it is categorized as High (systematic review/large surveillance/strong consensus), Moderate (consistent observational evidence with limitations), or Contested/Mixed (heterogeneous findings, causal uncertainty).

| Participation system                         | Documented harms (examples)                                                                                                                                                                          | Evidence and strength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Large social media participation by youths   | Associations with depressive symptoms, anxiety, sleep impacts; concerns that robust independent safety analyses are insufficient; risks heightened by adolescence and “almost constant” use patterns | US Surgeon General advisory emphasizes widespread use and cautions that safety evidence is insufficient for confident safety conclusions; it explicitly notes lack of “robust independent safety analyses.” (Moderate-to-High for “risk concern” framing; causal direction varies by study.)<br>Meta-analyses find associations between social media use, mental health, and sleep in young people, but literature is heterogeneous. (Moderate) Longitudinal work in <i>JAMA Network Open</i> reports that greater time on social media in early adolescence may contribute to increased depressive symptoms over time. (Moderate) |
| Online harassment participation environments | Harassment prevalence; escalation/intensification; differentiated impacts by group                                                                                                                   | Pew Research Center reports substantial prevalence of online harassment among US adults and notes intensification patterns. (High for prevalence)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



|                                                                          |                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Platform-mediated gig delivery work / last-mile delivery                 | Occupational safety risks (road crashes, injuries), psychosocial strain and precariousness driven by business model/incentives                                                                   | A 2025 systematic literature review specifically targets last-mile delivery workers' occupational risks and safety. (Moderate-to-High) Human Rights Watch reports survey findings that over one-third experienced at least one work-related car accident and around a quarter reported work-related injuries, framing safety risk as structural to platform work conditions. (Moderate)                                                                                                                      |
| Contact sports participation (repeated head impacts)                     | Increased concussion risk; potential long-term brain changes; association between repeated head impacts and CTE risk factors; ongoing scientific and clinical uncertainty about precise pathways | Centers for Disease Control and Prevention summarizes concussion/TBI risk framing and notes that repeated head impacts increase concussion chances and potential brain changes, while also clarifying limits of evidence about single/occasional concussions and CTE. (High for repeated impacts as risk, nuanced for CTE causal certainty) PubMed-indexed consensus criteria articles and major biology findings support plausible biological mechanisms and ongoing research expansion. (Moderate-to-High) |
| High-stakes platform design incentives (speed filter / risky engagement) | Design features can incentivize dangerous behavior; harms can be foreseeable given feature function and prior patterns                                                                           | US appellate decision allowed negligent design claims to proceed where the claim targeted product design rather than publishing third-party content. (Legal/empirical convergence; Moderate)                                                                                                                                                                                                                                                                                                                 |

Empirically, these domains share a critical structural feature: the organizer/platform typically holds superior telemetry about engagement, risk, incident rates, user reports, and design effects—precisely the inputs required for safety modeling and outcome disclosure. Ofcom guidance and the DSA explicitly treat such system characteristics and algorithmic factors as essential to risk assessment (which implies that the “foreseeability” baseline is often internal to the platform).

## Synthesis, remedies, and policy recommendations

## Liability outcomes and rationales: illustrative case and regulatory examples

The table below gives examples in which inadequate disclosure, risk governance failure, or design-driven foreseeability affected legal outcomes or institutional duties. (Not all are “standardized outcome disclosure” cases in the narrow sense; they are included because they show how law processes foreseeability + disclosure + institutional control.)

| Example                                                  | Jurisdiction | Participation system & alleged failure                                                 | Outcome (high level)                                                                                                                                                                     | Rationale linked to consent/disclosure/modeling                                                                                                                                                            |
|----------------------------------------------------------|--------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Canterbury v. Spence</i> (D.C. Cir. 1972)             | US           | Medical participation system; failure to disclose risk/alternatives adequately         | Court recognized patient autonomy as “informed exercise of and treated informed consent as requiring disclosure of material risks; endorsed objective “prudent person” causation framing | Consent is only meaningful choice”; duty arises from patient reliance and autonomy; risk significance depends on incidence and severity; causation handled via what a prudent person would do if informed. |
| <i>Montgomery v Lanarkshire Health Board</i> (UKSC 2015) | UK           | Clinical participation; failure to disclose material risks and reasonable alternatives | UK Supreme Court adopted a patient-centered material risk standard                                                                                                                       | The duty is framed around patient autonomy; reasonable care requires ensuring awareness of material risks and reasonable alternatives.                                                                     |

|                                                           |    |                                                                                              |                                                                                                                                        |                                                                                                                                                                              |
|-----------------------------------------------------------|----|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>House of Lords decision in Chester v Afshar</i> (2004) | UK | Clinical participation; failure to warn of small but serious risk                            | Court discussed altering conventional causation analysis in failure-to-warn context                                                    | Illustrates that consent failures can implicate autonomy interests in ways that reshape causation analysis, strengthening liability even when “the risk would exist anyway.” |
| US human-subjects research regulation (45 CFR 46.116)     | US | Research participation; inadequate key info/foreseeable risks; use of exculpatory releases   | Prohibits exculpatory language and mandates reasonable-person info + foreseeable risk disclosure                                       | In regulated research, this directly blocks institutional strategies to contract around negligence and anchors “informed” consent to reasonable-person material information. |
| Planet49 cookie consent press release (CJEU, 2019)        | EU | Data participation system; consent obtained via pre-ticked box                               | Official summary: pre-checked boxes are insufficient; consent must be active                                                           | Provides an EU example that formalistic consent mechanisms can fail where they do not reflect real, informed, affirmative authorization.                                     |
| Orange Romania consent press release (CJEU, 2020)         | EU | Data participation system; contract/tick-box consent for ID copy storage; friction to refuse | Official summary signals elevated standard for freely given, specific, informed consent and scrutiny of procedures that burden refusal | Shows that “consent” validity depends on real choice architecture and clarity, not just signatures.                                                                          |

|                                                                                  |    |                                                                                                                                                                                                     |                                                                                                                                           |                                                                                                                                                                                                                                                  |
|----------------------------------------------------------------------------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DSA systemic risk assessments (Regulation (EU) 2022/2065)                        | EU | Platform participation system; mandated annual systemic risk assessment and mitigation for very large platforms/search engines                                                                      | Creates affirmative duties to assess and mitigate systemic risks, including “foreseeable negative effects,” with transparency obligations | Turns “safety modeling” into a legal compliance obligation; failure implies regulatory liability and supplies a standard-of-care benchmark.                                                                                                      |
| UK Online Safety risk assessment guidance (Ofcom)                                | UK | Platform participation system; “suitable and sufficient” risk assessment; expectation to assess algorithmic systems, intended/unintended use; record-keeping; in some categories, publish summaries | Noncompliance can trigger enforcement and large penalties; governance expectations are spelled out                                        | Creates a standardized modeling and (for certain services) disclosure architecture; absence undermines claims that users “assumed” unknown risks and provides explicit regulatory breach hooks.                                                  |
| <i>Lemmon v Snap Inc.</i> (9th Cir. 2021)                                        | US | Platform participation + product design feature allegedly incentivizing dangerous speeding; defense asserted online intermediary immunity                                                           | Court held claim could proceed where it targeted negligent design, not publishing third-party content                                     | Demonstrates that “platform immunity” arguments are weaker when plaintiffs plead design-based foreseeability rather than publisher/speaker liability; supports shifting responsibility to design-controlling institutions.                       |
| National Football League concussion settlement appellate approval (3d Cir. 2016) | US | Participation system with long-documented head injury risks; litigation addressed knowledge, disclosure, and responsibility allocation                                                              | Appellate court affirmed settlement framework                                                                                             | Illustrates institutional settlement responses to structurally recurrent harms; while not a pure “consent invalidation” case, it reflects the legal system’s recognition of recurring harm patterns and institutional accountability mechanisms. |

## Flowchart: how disclosure absence changes consent analysis and liability posture

mermaid

Copy

flowchart TD

```
A[Participation system exists] --> B{Harms historically documented,\nstructurally recurrent, foreseeable?}
B -- No --> C[Ordinary risk baseline:\nstandard notice + general duty of care]
B -- Yes --> D[Heightened governance expectation:\ncontinuous risk assessment + monitoring]
D --> E{Standardized outcome disclosure\nand safety modeling present?}
E -- Yes --> F[Participant receives material risk info:\nprobability/severity + alternatives + safeguards]
F --> G{Harm occurs?}
G -- No --> H[System functioning;\nongoing oversight]
G -- Yes --> I[Liability analysis focuses on:\nreasonableness of controls + residual risk]
E -- No --> J[Information asymmetry persists:\nparticipant cannot meaningfully price risk]
J --> K{Is consent legally required\nfor lawfulness in this domain?}
K -- Yes --> L[Consent may be ineffective:\n"uninformed"/not freely given\n(or noncompliant consent process)]
L --> M[Institution/platform faces:\nregulatory breach + civil exposure]
K -- No --> N[Consent operates mainly as defense narrative:\nwaiver/assumption of risk]
N --> O[Defenses weakened:\nfailure-to-warn, deceptive design,\nnegligent governance theories strengthen]
M --> P[Remedies:\ninjunctions, fines, damages,\nstructural compliance orders]
O --> P
```

## Conclusions

A strong but qualified “yes.” In participation systems with documented, structurally recurrent, foreseeable harms, the absence of standardized outcome disclosure and safety modeling often:

1. Undermines validity of “informed consent” where consent is a legal prerequisite (notably in regulated research and data protection). In US human-subjects research, consent must provide reasonable-person information and foreseeable risk disclosure and cannot waive rights or release institutions from negligence. In EU data protection law, “informed” consent is mandatory and regulators and courts treat consent mechanisms as invalid when they are not genuinely affirmative and informed.

2. Weakens or defeats “participant-liability” defenses in less formal consent domains (consumer platform participation, recreational systems), especially when waivers conflict with public policy or statutory constraints (e.g., UK’s prohibition on excluding liability for death/personal injury from negligence; US public policy invalidation of certain releases).
3. Supports a duty/breach narrative against institutions/platforms by demonstrating that organizers did not take reasonable steps to identify and mitigate foreseeable harms—particularly where law expressly requires risk assessments and periodic review (DSA; UK online safety).
4. Shifts litigation posture even when proximate causation is contested. Where claims target system design and foreseeable behavioral incentives rather than third-party speech, US courts have sometimes allowed claims to proceed notwithstanding intermediary immunity arguments.

The key limitation is that “invalid consent → liability shifts” is not automatic. Liability still depends on (i) existence and scope of duty, (ii) breach (including what modeling/disclosure standard applies), (iii) causation, and (iv) defenses and statutory immunities. The absence of standardized disclosure/modeling is best understood as a multiplier: it turns structural foreseeability into stronger duty/breach proof and erodes defenses based on alleged voluntary risk assumption.

## **Policy recommendations**

Standardize safety modeling and outcome disclosure by risk tier. The DSA and UK online-safety regime provide templates: require periodic risk assessments keyed to service characteristics (including algorithms and business model), documented evidence inputs, and mandated review after significant changes. A parallel approach can be extended to other participation systems: high-risk tiers trigger stronger obligations (independent audits, public risk summaries, incident reporting).

Make “meaningful informed choice” a design requirement, not only a text requirement.

Regulator emphasis on deceptive design (“dark patterns”) suggests that disclosure must be

evaluated as a function of interface architecture and friction allocations, not merely as presence of legal text.

Create participant-facing outcome dashboards where organizers have telemetry. Ofcom's approach to requiring publication of risk assessment summaries for certain categories of services illustrates a feasible disclosure mechanism (risk levels, nature/severity of potential harm) while avoiding raw sensitive data exposure. In non-platform settings, analogs include standardized injury/adverse-event rates, complaint and enforcement metrics, and independent evaluations.

Align civil liability incentives with risk governance. The revised EU product liability regime explicitly incorporates software into "product" scope, reflecting a move toward treating digital systems as safety-relevant artifacts subject to no-fault liability principles. US product liability doctrine similarly recognizes strict liability foundations (e.g., Restatement §402A framing) that can, depending on classification as product vs service, discipline safety investment.

Protect vulnerable participants and reduce coercive participation conditions. Ethically and legally, power asymmetry (economic necessity; essential services; youth) undermines "choice" narratives, supporting heightened disclosure and safety requirements.

## **Open questions**

Standardization scope. Which outcomes must be disclosed to be "material" across heterogeneous systems (physical injury rates, mental health risks, economic loss distributions, exposure to illegal content), and what is the minimally sufficient histogram/summary? Ofcom's "risk profiles" model suggests one approach, but generalizing it raises measurement and privacy questions.

Metrics governance vs privacy. Safety modeling demands telemetry and monitoring, which can conflict with data minimization and privacy principles; designing outcome disclosure without incentivizing surveillance is a structural challenge. GDPR's centrality and DSA's research-access and documentation logic show both the need and the tension.

Civil enforcement pathways under platform safety regulation. DSA and UK Online Safety frameworks clearly increase regulatory accountability, but the extent to which they create or

strengthen private rights of action varies and remains a primary site of future litigation and legislative refinement.

Boundary between “product” and “service” for software platforms. EU product-liability modernization explicitly treats software as a product for no-fault liability purposes, potentially reshaping exposure landscapes for digital ecosystems; analogous classification in US law is more fragmented, affecting whether strict liability reaches platform-mediated harms.